How the Initial Estimate Is Certified

(Dated: October 15, 2026)

The finite-scale input $P_s(2304, 784), P_\tau(4608, 784) \geq 0.99$ of Ref. [1] is established without direct tensor contraction at that scale. Ref. [3] certifies a pair of bounds at length 36 and $\beta = 49/4$, which are then propagated to $(2304, 784)$ through six scale doublings with decoupled defects. The underlying verification reduces to three closed tensor networks: twisted rings of at most twelve sites, symmetry-projected rings, and an integer matrix-product trial state of bond dimension 26 on 72 sites.

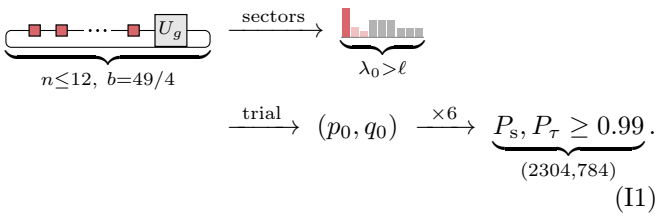
Appendix I: Architecture of the Certificate

The proof of Ref. [1] takes one numerical input: the spatial and thermal purities at length 2304 and inverse temperature 784 are at least 0.99. Computing them directly would require partition functions of rings of 4608 spins, far beyond any computer. Ref. [3] certifies them instead from rings of at most twelve sites, and this note explains how.

The idea is to start at a much smaller scale and let the doubling recurrence carry the bounds up. At length 36 and inverse temperature 49/4, Ref. [3] certifies upper bounds p_0 and q_0 on the spatial and thermal defects. Six doublings take $(36, 49/4)$ to $(36 \cdot 2^6, \frac{49}{4} \cdot 2^6) = (2304, 784)$, and along the way both defects fall below $1/100$.

We write b for an inverse temperature in this note, mainly the certificate value $b = 49/4$, and X_b for the spatial transfer operator of Ref. [1] at b. Its eigenvalues $\lambda_i(b)$ are real, and $Z_n(b) = \sum_i \lambda_i^n$. Following Ref. [3], the Hamiltonian is shifted by $\kappa = 700741/500000$ per site, $H_n \to H_n + \kappa n$. This κ is set by the matrix-product trial state of Sec. K: it is chosen so that $-\kappa$ lies just above the trial state's energy per site, so the trial state certifies $Z_{72}(b) > 1$. After the shift the leading eigenvalue of X_b is close to one. The shift cancels from every purity.

The certificate proceeds in four steps, from twisted partition functions $\mathrm{Tr}(X_b^n U_g)$ to a leading-eigenvalue lower bound $\lambda_0 > \ell$, starting defects (p_0, q_0) at $(36, 49/4)$, and six doublings to the starting estimate of Ref. [1]:



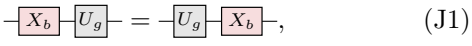
(I1)

- *Twisted partition functions* (Sec. J): certified values of $Z_n(b, g)$ for $4 \leq n \leq 12$, $b = 49/4$, and two twists g.
- *Sectors and filters* (Sec. K): bounds on every eigenvalue of X_b, sector by sector. A trial vector then forces one eigenvalue above $\ell = 0.999$.
- *Initial defects* (Sec. K): $1 - P_s(36, b) \leq p_0$ and $1 - P_\tau(72, b) \leq q_0$.
- *Six doublings* (Sec. L): the coupled update reaches $(2304, 784)$.

Every number is an exact rational or an integer; floating-point arithmetic is used nowhere in the proof.

Appendix J: Twisted boundary conditions

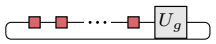
Twists and symmetries. Untwisted traces $Z_n(b) = \sum_i \lambda_i(b)^n$ sum over all spin-rotation sectors at once; separating the D_2-invariant sector from the three equivalent nontrivial sectors (the eigenvalue lists J and N of Sec. K) requires twisted traces. A proper signed permutation g of the Cartesian axes acts on histories by U_g, which relabels events and multiplies by their signs. Rotating every matrix inside a site trace shows that U_g slides through a transfer vertex,



(J1)

so a single U_g may be placed on any virtual bond of the ring. On the physical side it conjugates one end of a single bond, the seam:

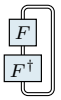
$$Z_n(b, g) = \mathrm{Tr}\, X_b^n U_g = \quad = \mathrm{Tr}\, e^{-bH_n(g)}. \tag{J2}$$



The certificate needs $g = 1$ and the π rotation $g = P$.

Purification. The physical side is computed as a squared Frobenius norm, which splits the imaginary-time circle at half its length,

$$Z_n(b, g) = \quad = \|F\|_F^2, \qquad F = e^{-bH/2}, \tag{J3}$$



so that only columns Fe_r are needed, one basis state e_r at a time.

Uniformization. Shift and scale H so that $R = 1 - (\frac{3}{2}n + H)/16$ is a contraction. Since $8b = 98$ at $b = 49/4$, uniformization (Jensen's method) writes F, up to the scalar $e^{-b(\kappa - 3/2)n/2}$, as a Poisson mixture of powers of R:

$$F \propto e^{98(R-1)} = \sum_{j \geq 0} e^{-98} \frac{98^j}{j!} \quad , \tag{J4}$$



with j factors of R in the jth term. This is the discrete counterpart of the event expansion behind the thermal

MPO: the events of W_β arrive as a Poisson process, and here each factor of R is one event.

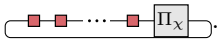
The sum is cut at $j = 280$. The truncation error is below 2^{-84} in operator norm, because $\|R\| \le 1$ [4]. The truncated sum is evaluated by Horner's rule on integer vectors scaled by $Q = 2^{55}$, taking the floor after every multiplication by R. Each multiplication adds at most $\sqrt{\dim} + 1$ to the Euclidean error, and the contraction carries earlier errors forward unamplified. Each column is therefore known to within $281(\sqrt{3^n} + 2)$ in units of $1/Q$.

Orbits. The trace in Eq. (J3) needs one column per basis word. Cyclic shifts, reversal and global spin flip permute the columns without changing their norms. For the twisted ring, a shift moves the seam, and a diagonal sign gauge moves it back. Only one representative per orbit is computed, weighted by the orbit size. The results are eighteen integers $W_{n,g}$ of about 30 digits. Turning them into rational enclosures gives the table of $Z_n(49/4, g)$, each value to within $\pm 3 \times 10^{-5}$. As an independent sanity check, not part of the proof, floating-point exact diagonalization reproduces all ten centers for $4 \le n \le 8$ to within 10^{-6}.

Appendix K: From partition functions to the transfer spectrum

Sector projectors. The four twists $g \in D_2 = \{1, P_x, P_y, P_z\}$ commute with X_b, so their character sums are projectors that can be inserted into the ring:

$$\Pi_\chi = \tfrac{1}{4} \textstyle\sum_{g \in D_2} \chi(g) U_g, \qquad \operatorname{Tr} X_b^n \Pi_\chi = \tag{K1}$$



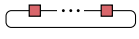
A cyclic rotation of the axes permutes the three nontrivial characters, so those three sectors carry identical eigenvalue lists, called N. Write J for the list of the invariant sector. Inverting the characters gives both lists from the two twists alone:

$$J_n = \tfrac{1}{4}\big[Z_n + 3Z_n(P)\big], \qquad N_n = \tfrac{1}{4}\big[Z_n - Z_n(P)\big], \tag{K2}$$

with $Z_n = J_n + 3N_n$. The coefficients sum to at most one in absolute value, so each sector moment inherits the 3×10^{-5} error. This holds for odd n too; odd moments carry sign information.

Polynomial filters (the term of Ref. [3]). For any polynomial $F \ge 0$ on $\mathbb{R}$, the closed network $\operatorname{Tr} F(X_b)$ is a finite combination of ring partition functions:

$$\sum_i F(\lambda_i) = \sum_j f_j \; \le \; B. \tag{K3}$$



Every term on the left is nonnegative, so no single eigenvalue can have $F(\lambda) > B$. Two degree-12 filters of the form $x^4 Q(x)^2$ give

$$|\lambda| < v_J = 1.00139 \text{ in } J, \qquad |\lambda| < v_N = 0.872 \text{ in } N. \tag{K4}$$

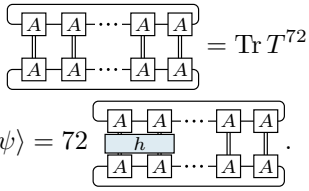
The filter needs $F \ge 0$ on the whole real line, not only on $[0, \infty)$, because X_b is self-adjoint but not positive. Reflection positivity is what makes the eigenvalues real [2].

Capped masses (also Ref. [3]'s term; a convexity bound). The twelfth moments are bounded by $m_J = 1.0657205$ and $m_N = 0.2783155$. Weights $t_i \le h$ with total mass at most m have their rth powers maximized by filling bars to the cap,  :

$$\sum_i t_i^r \le \mathcal{C}(m, h, r) = qh^r + (m - qh)^r, \qquad q = \lfloor m/h \rfloor. \tag{K5}$$

The trial vector. One matrix-product vector on 72 sites, with real integer tensors A_s of bond dimension 26, has energy per site below $-\kappa$. Its norm and energy are two double-layer rings,

$$\langle \psi | \psi \rangle = \; = \operatorname{Tr} T^{72}, \qquad \langle \psi | H | \psi \rangle = 72 \; . \tag{K6}$$



The virtual space carries four spin-$\frac{1}{2}$, three spin-$\frac{3}{2}$ and one spin-$\frac{5}{2}$ multiplets, $4{\cdot}2{+}3{\cdot}4{+}6 = 26$. The contraction splits into blocks of dimension at most 162, evaluated in exact integers. The outcome, $E_0(72) < -72\kappa$, means that one Boltzmann factor of $Z_{72}(b)$ exceeds one, so

$$Z_{72}(b) > 1 \quad \text{for every } b > 0. \tag{K7}$$

A dominant eigenvalue. Suppose every eigenvalue in J had $|\lambda| \le \ell = 0.999$. The capped-mass bound applied to J and to the three copies of N would give $Z_{72}(49/4) \le 0.931 < 1$, contradicting Eq. (K7). Hence some $\lambda_0 \in J$ has $|\lambda_0| > \ell$: the spectrum looks like  , with one tall bar.

Removing λ_0 leaves the following bound on the kth moment of the rest:

$$R_A(k) = 3\,\mathcal{C}\big(m_N, v_N^{12}, \tfrac{k}{12}\big) + \big(m_J - \ell^{12}\big)^{k/12}. \tag{K8}$$

Initial defects. Using $Z_{72}(b) \ge \lambda_0^{72} > \ell^{72}$ in $P_{\mathrm{s}}(36, b) = Z_{72}(b)/Z_{36}(b)^2$ and $Z_{72}(2b) > 1$ [Eq. (K7)] in $P_\tau(72, b) = Z_{72}(2b)/Z_{72}(b)^2$ bounds both purities at the base scale, giving $(p_0, q_0) \simeq (0.0479, 0.1815)$:

$$P_{\mathrm{s}}(36, b) = \; \ge \Big(1 + \frac{R_A(36)}{\ell^{36}}\Big)^{-2} = 1 - p_0,$$

$$P_\tau(72, b) = \frac{Z_{72}(2b)}{Z_{72}(b)^2} \ge \big(v_J^{72} + R_A(72)\big)^{-2} = 1 - q_0. \tag{K9}$$

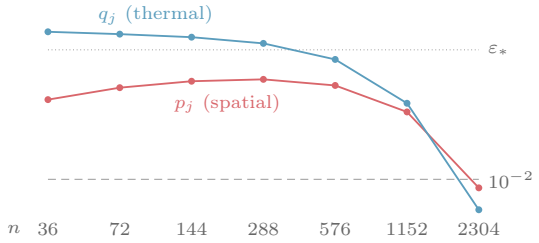


Appendix L: Six doublings

The thermal defect q_0 exceeds the diagonal threshold $\varepsilon_* \simeq 0.127$ of Ref. [2]. A single common ε would therefore not start the iteration. Instead, Moves 1–4 of Ref. [1]

track the two defects separately; with $f(z) = z^2/[2(1-z)^2]$ and $\lceil\cdot\rceil_7$ rounding up to a multiple of 10^{-7},

$$\begin{aligned} p_{j+1} &= \lceil f\big(1-(1-p_j)^2(1-q_j)\big)\rceil_7, \\ q_{j+1} &= \lceil f\big(1-(1-q_j)^2(1-p_{j+1})\big)\rceil_7. \end{aligned} \tag{L1}$$



The figure plots the defects on a logarithmic scale against the length $n_j = 36 \cdot 2^j$; the inverse temperature doubles alongside.

- The thermal defect falls from the first step.
- The spatial defect first *rises*, to 0.071 at $j = 3$: at small n the coupling identity costs more than squaring gains.
- Both then collapse together and cross 10^{-2} at the last step,

$$\begin{aligned} 1 - P_{\mathrm{s}}(2304, 784) &\le 0.0084513, \\ 1 - P_{\tau}(4608, 784) &\le 0.0054986. \end{aligned} \tag{L2}$$

This recovers the initial estimate of Ref. [1] (Proposition 5.5 of Ref. [3]); re-evaluating the recurrence from the rational definitions of p_0 and q_0 confirms agreement digit for digit.

The weaker initial estimate and temperature lifting. Ref. [3] also certifies $P_{\mathrm{s}}(60, 105/4)$, $P_{\tau}(120, 105/4) > 7/8$ from a table at $b = 21/2$ for $n \le 10$, three twists, the $N = 120$ trial vector, and $\sum_m z_m^{5/2} \le (\sum_m z_m)^{5/2}$ for the Boltzmann factors $z_m = e^{-bE_m}$, giving $\Delta_L > \frac{4}{105}\log\frac{80}{79}$ for even $L \ge 60$. Checking the finite prefix intervals $[n_j, 2n_j]$ via $P_{\tau}(L, b_j) \ge (1-p_j)^2(1-q_j)$ and sharpening (v_J, v_N, ℓ) in Eq. (K9) to $(1.001351, 0.871256, 0.999998)$ already extends the $(36, 49/4)$ certificate to all even $L \ge 36$ with $\Delta_L > 0.01697$ (`models/spin1/experiments/`). Raising the inverse temperature directly on the $b_0 = 49/4$ sector moments before reducing to two purities gives a 19.1× larger bound than Ref. [1]: for even $12 \le k \le 60$ and $r \ge 1$, Eq. (K5) bounds $Z_k(b_0, P) \le J_k \le \mathcal{C}(m_J, v_J^{12}, k/12)$ and $Z_k(b_0) \le M_k := \mathcal{C}(m_J, v_J^{12}, k/12) + 3\,\mathcal{C}(m_N, v_N^{12}, k/12)$. Raising the positive Boltzmann factors of the self-adjoint twisted Hamiltonians to power r gives $Z_k(rb_0) \le M_k^r =: D$ and $Z_k(rb_0, P) \le \mathcal{C}(m_J, v_J^{12}, k/12)^r =: E$, so at $b = rb_0$ the spatial weights $u_i = |\lambda_i(b)|^k$ satisfy $\sum_i u_i \le D$, J-sector mass at most $\tau = (D + 3E)/4$, and $\sum_i u_i^{2h} = Z_{120}(b) > 1$ $(h = 60/k)$. Because each N eigenvalue has multiplicity three, any $t > D/3$ with $Dt^{2h-1} < 1$ forces $u_0 \in J$ with $u_0 > t$, leaving residual sth moment at most

$$R_s(t) = \max\left\{3\Big(\frac{D-t}{3}\Big)^s,\ (\tau - t)^s + 3\Big(\frac{D-\tau}{3}\Big)^s\right\}, \tag{L3}$$

upgraded whenever $(t')^{2h} + R_{2h}(t) < 1$. For the physical Boltzmann factors $z_m = e^{-b_0 E_m(120)}$, $Z_{120}(b_0) \le M_{120}$ and $z_0 > 1$ give $\sum_{m>0}(z_m/z_0)^r \le (M_{120} - 1)^r$, yielding

$$\begin{aligned} P_{\mathrm{s}}(60, rb_0) &\ge \Big(1 + \frac{R_h(t)}{t^h}\Big)^{-2}, \\ P_{\tau}(120, rb_0) &\ge \big(1 + (M_{120} - 1)^r\big)^{-2}. \end{aligned} \tag{L4}$$

Combined with the $(36, 49/4)$ prefix on $36 \le L \le 72$, Eq. (L4) certifies $\Delta_L > 0.07045 > 7/100$ at $(k, r) = (30, 2)$ and $\Delta_L > 0.07290$ at $(28, 17/8)$ for every even $L \ge 36$. With a single untwisted bound $Z_n(b_0) \le U$ and trial $Z_{2hn}(b_0) > 1$ $(h, r \ge 1)$, the same argument with $D = U^r$ and $t > D/2$ gives $P_{\mathrm{s}}(hn, rb_0) \ge [1 + ((D-t)/t)^h]^{-2}$ and $P_{\tau}(2hn, rb_0) \ge [1 + (U^{2h} - 1)^r]^{-2}$ whenever $t^{2h} + (D-t)^{2h} < 1$ (`models/spin2/scripts/one_trace_target.jl`).

Appendix M: Model-Dependent vs Universal Components

Each ingredient is a closed network, and only a few of them depend on the model.

- *Model independent:* the uniformization of Eq. (J4), the orbit reduction, the filters of Eq. (K3), the capped masses, and the doubling of Eq. (L1).
- *Model specific:*
  - the symmetry group that splits the spectrum into sectors, which for spin 1 makes N appear three times;
  - one good trial vector;
  - the sizes at which the filters isolate one eigenvalue.

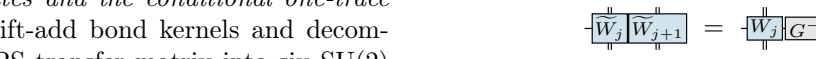
Smaller seed inputs for formal verification. Five studies in `models/lean_seeds` examine how to discharge the spin-1 seed inside the Lean 4 kernel (`lean/SeedKernel/`). First, even lengths alone suffice: $n \in \{4, 6, 8, 10, 12\}$ with the two real twists $g \in \{1, P\}$ at $b = 49/4 \to 49/2$ and trace error 10^{-3} certifies $\Delta_L > 0.039036$ for even $L \ge 60$, while $n \in \{6, 8, 10\}$ with twists $1, P, C$ (C the $2\pi/3$ axis permutation) at $b = 21/2$ and error 3×10^{-4} certifies $\Delta_L > 0.005966$ while cutting the orbit-trace dimension by 8.01×. Second, replacing Poisson uniformization by low-energy trial eigenstates ⌊ (as done unconditionally for the two-state Ising seed in Ref. [2]) and a half-chain comparison operator $H_n \succeq H_0 = A_{n/2} \otimes I + I \otimes A_{n/2} + 2cI$ (certified by exact rational 81×81 block $\mathrm{LDL}^{\mathsf{T}}$ factorization) checks $n = 12$ packed `Nat` matrix-vector products in 9.6–26 s. Unlike the Ising seed, however, spin 1

at $n = 12, b = 49/4$ has 26 ($g = 1$) and 34 ($g = P$) eigenstates below the tail cutoff, and the $S^z = 0$ comparison count (49 with a 32-bin tail vs. 8 physical levels) does not yet close without homotopy continuation, so a kernel-checked spin-1 seed remains open.

Spin-2 trial certificates and the conditional one-trace target. For spin 1, shift-add bond kernels and decomposing the $D = 26$ MPS transfer matrix into six SU(2) multiplicity blocks (largest dimension 56 vs. 162 in S^z) speed up the exact length-120 trial contraction by 360× (`models/spin1`). For spin 2 ($\boldsymbol{S}^2 = 6I$, $5^{12} \simeq 2.4 \times 10^8$ states, $\xi \approx 49.5$), the reflection-positive bond form and the bootstrap recurrence carry over unchanged, and SU(2)-invariant MPS trials certify $E_0(N)/N < -4.7595$ ($D = 24$, integer) and < -4.7612463 ($D = 295$, Arb; reduced by Schur's lemma and the π-rotation involution from $295^2 = 87{,}025$ to blocks of size at most 978) up to $N = 1024$ (`models/spin2`). By the untwisted one-trace bound above at $(n, b_0, h, r) = (256, 64, 2, 3)$ with shift $\kappa = 4.7612463$, a single thermal upper bound $Z_{256}(64) \le 1.08$ would certify $\Delta_L > 0.003019$ for even $L \ge 512$ (whereas the $(128, 32)$ target at $\kappa = 4.76122$ is ruled out by the $N = 128$ trial lower bound $Z_{128}(32) > 1.1967$). Certifying $Z_{256}(64) \le 1.08$ remains open.

Certified thermal MPO compression. Two routes in `models/spin2` bound $Z_N(b)$ from above before MPO compression (`models/spin2/IDEAS.md`): shifting $K = H_N + NI \succeq 0$ ($\|K\| \le 5N$), a degree-d Taylor majorant $Q_d(K) = \sum_{j=0}^{d}(-\delta K)^j/j! + rI \succeq e^{-\delta K}$ ($\delta = b/(2m)$, $m = 2^p$) gives $Z_N(b) \le e^{(1-\kappa)bN}\,\mathrm{Tr}\,Q_d(K)^{2m}$, where the extensive norm $\delta\|K\| \le 5N\delta$ requires $p = 28$ squarings at $N = 256, b = 64$ for $d = 2$, or $p = 21$ for $d = 4$ (factored into a positive rational sum of squares of quadratics via the LDL^T decomposition of the Laguerre moment matrix, $P_4(x) = (1 + \frac{x}{2} + \frac{x^2}{12})^2 + \frac{1}{12}(x + \frac{x^2}{2})^2 + \frac{1}{72}(x^2)^2$); alternatively, the Araki–Lieb–Thirring inequality for even and odd bond layers $H_N = A + B$ gives $Z_N(b) \le \mathrm{Tr}[(e^{-bA/m}e^{-bB/m})^m]$ with step error governed by the local bond norm rather than N. To certify MPO compressions along repeated squarings, `CertifiedGaps.CertifiedMPO` (`julia/src/CertifiedMPO/`) uses two tensor-network mechanisms:

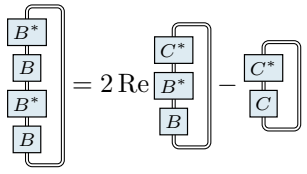
- *Rational matrix gauges.* Diagonal virtual-bond rescalings leave the local core-norm product $\prod_j \|W_j\|_{\mathrm{local}}$ at $N = 256$ a factor of 1.66×10^{75} above the analytic upper bound a_0 on $\|Q_2/5^{N/2}\|$, whereas invertible rational gauge matrices $\widetilde{W}_j = G_j W_j G_{j+1}^{-1}$ derived from the Schmidt decomposition cancel on every internal virtual bond,



(M1)

  bringing the Cholesky-certified product $\prod_j \|\widetilde{W}_j\|_{\mathrm{local}}$ within $1.000000701\,a_0$.

- *Three-layer trace correction and backward error propagation.* When a squared step $A = B^*B$ is compressed to C with $\|A - C\|_{\mathrm{F}} \le \eta$, the identity



(M2)

$$+ \|B^*B - C\|_{\mathrm{F}}^2$$

  replaces the four-layer trace $\mathrm{Tr}(A^2)$ and linear triangle error $(\|C\|_{\mathrm{F}} + \eta)^2$ by a three-layer signed overlap with quadratic residual η^2 (297× tighter single-step excess at $D = 32$). Across a multi-step flow, propagating approximate backward adjoints (`AdjointTrace.jl`) retains the signed linear errors across steps (1307× narrower $N = 16$ four-step enclosure), while backward Schatten bounds (`SchattenBridge.jl`) track $\|B_j\|_{p_j}$ with $p_j = 2^{k+1-j}$ to use operator-norm residuals at early steps and Hilbert–Schmidt residuals near the final trace. These certificates pass replay on 4- and 16-site flows and the first two squarings at $N = 256$ ($b = 2^{-20}$), leaving 26 squarings to $b = 64$ open.

[1] Companion note, *A diagrammatic proof of the Haldane gap via spacetime tensor networks*, `notes/thermal-mpo-gap-note.tex` (2026).

[2] Companion note, *Scope of the spacetime bootstrap*, `notes/thermal-mpo-gap-scope.tex` (2026).

[3] OpenAI, *The periodic spin-one Haldane gap*, manuscript dated September 24, 2026, Secs. 3–5 and Appendices A–C.

[4] B. L. Fox and P. W. Glynn, Computing Poisson probabilities, Commun. ACM **31**, 440 (1988).